Hossein Aqasizade

Website | GitHub | LinkedIn | Gmail

Graduate Research Assistant, Department of Computer Engineering,
University of Mazandaran, Babolsar, Iran

Educational Background

Master Degree (2020 - 2023)

M.Sc., Computer Engineering, Computer Networks
Department of Computer Engineering, University of Mazandaran, Babolsar, Iran
Thesis: Combining Virtual Machines and Containers in a Cluster
Supervised by: Dr. Ehsan Ataie
Total GPA: 3.88/4 (17.92/20 in Iranian Scale)

Bachelor Degree (2016 - 2020)

B.Sc., Computer Engineering, Information Technology
Department of Computer Engineering, Azad University, Mazandaran, Iran
Final Project: Implemented management system with virtualization
Advised by: Dr. Habib Esmaeelzadeh Rostam
Total GPA: 3/4 (14.49/20 in Iranian Scale)

Experiences

Teaching Assistant

- Cloud Computing (for Master Students), Supervised by Dr. Ehsan Ataie (Fall 2022)
  Training (Kubernetes, Docker, Prometheus, and Grafana), and Project Management
- Cloud Computing (for Bachelor Students), Supervised by Dr. Ehsan Ataie (Fall 2022)
  Training (Kubernetes, Docker, Prometheus, and Grafana), and Project Management
- Performance Evaluation (for Master Students), Supervised by Dr. Ehsan Ataie (Fall 2021)
  Training, Exercise Definition, and Project Management

Research Assistant

- Performance Analysis of Serverless Platforms, Supervised by Dr. Ehsan Ataie (Nov 2022 - Aug 2023)
  As team leader, I focus on evaluating serverless platforms like Open-FaaS, Fission, Nuclio, Knative, and OpenWhisk. Our project aims to assess their performance in diverse Kubernetes setups and conditions.
- IT Support Specialist, Almas, Ghaemshahr county, Mazandaran, Iran (Jan 2017 - Feb 2020)

Journal Papers

- **Performance Analysis of Containers Running on Top of Virtual Machines (Submitted–2023)**
  Description: Investigating the performance implications of running containers within virtualized environments.
  Skills Developed: Proficiency in containerization, virtualization, and performance testing methodologies.
  Journal: Cluster Computing
  Authors: **Hossein Aqasizade**, Ehsan Ataie, Mostafa Bastam
- **Optimizing Container Performance in Kubernetes: A Study with OpenFaaS on Virtual Machines (In preparation)**
  Description: Studying the performance of serverless functions with OpenFaaS and MySQL on Kubernetes in virtualized environments, including Xen VMs in a CloudLab
  Skills Developed: Proficient in Kubernetes, Xen virtualization, OpenFaaS, MySQL, monitoring tools, and benchmarks.
- **A Comparative Study of Virtualization: Heavyweight Virtualization or Lightweight Virtualization? (Submitted–2023)**
  Description: Conducting a comparative analysis to evaluate the performance and resource utilization of heavyweight virtualization and lightweight virtualization techniques.
  Skills Developed: Proficiency in virtualization, experimental design, data analysis, and academic writing.
  Journal: The CSI Journal on Computing Science and Information Technology
  Authors: **Hossein Aqasizade**, Ehsan Ataie, Mostafa Bastam

- Performance Evaluation of Kubernetes Deployments Approaches with OpenFaaS Integration (In preparation)

  Description: Investigating the performance implications of integrating OpenFaaS with different deployment approaches in Kubernetes clusters.
  Skills Developed: Proficiency in Kubernetes orchestration, containerization technologies, serverless computing with OpenFaaS, experimental design, and data analysis.
  Authors: **Hossein Aqasizade**, Ehsan Ataie, Mostafa Bastam

Research Interests

- Serverless Computing
- Machine Learning
- Federated Learning
- High Performance Computing
- Parallel Computing
- Virtualization

Skills

- Kubernetes (show certificate)
- Docker (show certificate)
- LPIC 1 and LPIC 2
- LPIC 304 (show certificate)
- Python Advanced (show certificate)
- AWS ECS (show certificate)
- CloudLab
- NetConf, NetFlow, NetSNMP, NetXMS
- Adaptability and Problem-Solving
- Monitoring Tools (Prometheus, Grafana, and Zabbix)
- Microsoft Certificate (70-698, 70-410, and 70-411)
- Benchmarking Tools
- HPC (High Performance Computing)
- Virtualization (Xen, ESXi, KVM, and ...)
- CompTIA Network+
- Serverless Platforms (OpenFaaS, Knative, Fission, and Nuclio)
- HTML, CSS, JS
- Leadership and Teamwork

Educational Projects

- Implemented **Genetic Algorithm** for feature selection in Mobile Price classification using **Python**.
- Developed a **Python-based model** for categorizing Dry Beans, utilizing a **machine learning** framework.
- Configured **NetSNMP** Configuration for Network Monitoring.
- Configured **NetConf** Configuration for Network Device Management.
- Configured **NetFlow** Configuration for Traffic Analysis.
- Configured **NetXMS** Configuration for Network Monitoring.
- Implemented and simulated cloud computing environments using **CloudSim** toolkit in **Eclipse** IDE.
- Designed and deployed a queuing system simulation **(M/M/1/K)** using **Python**.
- Developed a Python program for a **graph-based access control system** using **Python**.

IELTS Score

- Will be taken in November 2023.

Professional References

- Dr. Ehsan Ataie (Supervisor)

  Assistant Professor, Department of Computer Engineering, University of Mazandaran, Babolsar, Iran
  E-mail: ataie@umz.ac.ir
- Dr. Mostafa Bastam (Advisor)

  Assistant Professor, Department of Computer Engineering, University of Mazandaran, Babolsar, Iran
  E-mail: bastam@umz.ac.ir
- Dr. Payam Mahmoudi-Nasr

  Assistant Professor, Department of Computer Engineering, University of Mazandaran, Babolsar, Iran
  E-mail: p.mahmoudi@umz.ac.ir